Testing the Generalization Power of Neural Network Models Across NLI Benchmarks

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Abstract

Neural network models have been very successful in natural language inference, with the best models reaching 90% accuracy in some benchmarks. However, the success of these models turns out to be largely benchmark specific. We show that models trained on a natural language inference dataset drawn from one benchmark fail to perform well in others, even if the notion of inference assumed in these benchmarks is the same or similar. We train six high performing neural network models on different datasets and show that each one of these has problems of generalizing when we replace the original test set with a test set taken from another corpus designed for the same task. In light of these results, we argue that most of the current neural network models are not able to generalize well in the task of natural language inference. We find that using large pre-trained language models helps with transfer learning when the datasets are similar enough. Our results also highlight that the current NLI datasets do not cover the different nuances of inference extensively enough.

1 Introduction

Natural Language Inference (NLI) has attracted considerable interest in the NLP community and, recently, a large number of neural network-based systems have been proposed to deal with the task. One can attempt a rough categorization of these systems into: a) sentence encoding systems, and b) other neural network systems. Both of them have been very successful, with the state of the art on the SNLI and MultiNLI datasets being 90.4%, which is our baseline with BERT (Devlin et al., 2019), and 86.7% (Devlin et al., 2019) respectively. However, a big question with respect to these systems is their ability to generalize outside the specific datasets they are trained and tested on. Recently, Glockner et al. (2018) have shown that state-of-the-art NLI systems break considerably easily when, instead of tested on the original SNLI test set, they are tested on a test set which is constructed by taking premises from the training set and creating several hypotheses from them by changing at most one word within the premise. The results show a very significant drop in accuracy for three of the four systems. The system that was more difficult to break and had the least loss in accuracy was the system by Chen et al. (2018) which utilizes external knowledge taken from WordNet (Miller, 1995).

In this paper we show that NLI systems that have been very successful in specific NLI benchmarks, fail to generalize when trained on a specific NLI dataset and then these trained models are tested across test sets taken from different NLI benchmarks. The results we get are in line with Glockner et al. (2018), showing that the generalization capability of the individual NLI systems is very limited, but, what is more, they further show the only system that was less prone to breaking in Glockner et al. (2018), breaks too in the experiments we have conducted.

We train six different state-of-the-art models on three different NLI datasets and test these trained models on an NLI test set taken from another dataset designed for the same NLI task, namely for the task to identify for sentence pairs in the dataset if one sentence entails the other one, if they are in contradiction with each other or if they are neutral with respect to inferential relationship.

One would expect that if a model learns to correctly identify inferential relationships in one dataset, then it would also be able to do so in another dataset designed for the same task. Furthermore, two of the datasets, SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2018), have been constructed using the same crowdsourcing approach and annotation instructions (Williams

et al., 2018), leading to datasets with the same or at least very similar definition of entailment. It is therefore reasonable to expect that transfer learning between these datasets is possible. As SICK (Marelli et al., 2014) dataset has been machine-constructed, a bigger difference in performance is expected.

In this paper we show that, contrary to our expectations, most models fail to generalize across the different datasets. However, our experiments also show that BERT (Devlin et al., 2019) performs much better than the other models in experiments between SNLI and MultiNLI. Nevertheless, even BERT fails when testing on SICK. In addition to the negative results, our experiments further highlight the power of pre-trained language models, like BERT, in NLI.

The negative results of this paper are significant for the NLP research community as well as to NLP practice as we would like our best models to not only to be able to perform well in a specific benchmark dataset, but rather capture the more general phenomenon this dataset is designed for. The main contribution of this paper is that it shows that most of the best performing neural network models for NLI fail in this regard. The second, and equally important, contribution is that our results highlight that the current NLI datasets do not capture the nuances of NLI extensively enough.

2 Related Work

The ability of NLI systems to generalize and related skepticism has been raised in a number of recent papers. Glockner et al. (2018) show that the generalization capabilities of state-of-the-art NLI systems, in cases where some kind of external lexical knowledge is needed, drops dramatically when the SNLI test set is replaced by a test set where the premise and the hypothesis are otherwise identical except for at most one word. The results show a very significant drop in accuracy. Kang et al. (2018) recognize the generalization problem that comes with training on datasets like SNLI, which tend to be *homogeneous* and with little linguistic variation. In this context, they propose to better train NLI models by making use of adversarial examples.

Multiple papers have reported hidden bias and annotation artifacts in the popular NLI datasets SNLI and MultiNLI allowing classification based on the hypothesis sentences alone (Tsuchiya, 2018; Gururangan et al., 2018; Poliak et al., 2018).

Wang et al. (2018) evaluate the robustness of NLI models using datasets where label preserving swapping operations have been applied, reporting significant performance drops compared to the results with the original dataset. In these experiments, like in the BreakingNLI experiment, the systems that seem to be performing the better, i.e. less prone to breaking, are the ones where some kind of external knowledge is used by the model (KIM by Chen et al., 2018 is one of those systems).

On a theoretical and methodological level, there is discussion on the nature of various NLI datasets, as well as the definition of what counts as NLI and what does not. For example, Chatzikyriakidis et al. (2017); Bernardy and Chatzikyriakidis (2019) present an overview of the most standard datasets for NLI and show that the definitions of inference in each of them are actually quite different, capturing only fragments of what seems to be a more general phenomenon.

Bowman et al. (2015) show that a simple LSTM model trained on the SNLI data fails when tested on SICK. However, their experiment is limited to this single architecture and dataset pair. Williams et al. (2018) show that different models that perform well on SNLI have lower accuracy on MultiNLI. However in their experiments they did not systematically test transfer learning between the two datasets, but instead used separate systems where the training and test data were drawn from the same corpora.[1]

3 Experimental Setup

In this section we describe the datasets and model architectures included in the experiments.

3.1 Data

We chose three different datasets for the experiments: SNLI, MultiNLI and SICK. All of them have been designed for NLI involving three-way classification with the labels *entailment*, *neutral* and *contradiction*. We did not include any datasets with two-way classification, e.g. SciTail (Khot et al., 2018). As SICK is a relatively small dataset with approximately only 10k sentence pairs, we did not use it as training data in any experiment.

[1] To be more precise, Williams et al. (2018) tested some transfer across datasets, but only between MultiNLI and SNLI.

Train data	Test data	Size of the training set	Size of the test set
SNLI	**SNLI**	550,152	10,000
SNLI	MultiNLI	550,152	20,000
SNLI	SICK	550,152	9,840
MultiNLI	**MultiNLI**	392,702	20,000
MultiNLI	SNLI	392,702	10,000
MultiNLI	SICK	392,702	9,840
SNLI + MultiNLI	**SNLI**	942,854	10,000
SNLI + MultiNLI	SICK	942,854	9,840

Table 1: Dataset combinations used in the experiments. The rows in bold are baseline experiments, where the test data comes from the same benchmark as the training and development data.

We also trained the models with a combined SNLI + MultiNLI training set.

For all the datasets we report the baseline performance where the training and test data are drawn from the same corpus. We then take these trained models and test them on a test set taken from another NLI corpus. For the case where the models are trained with SNLI + MultiNLI we report the baseline using the SNLI test data.[2] All the experimental combinations are listed in Table 1. Examples from the selected datasets are provided in Table 2. To be more precise, we vary three things: training dataset, model and testing dataset. We should qualify this though, since the three datasets we look at, can also be grouped by text domain/genre and type of data collection, with MultiNLI and SNLI using the same data collection style, and SNLI and SICK using roughly the same domain/genre. Hopefully, our set up will let us determine which of these factors matters the most.

We describe the source datasets in more detail below.

SNLI

The Stanford Natural Language Inference (SNLI) corpus (Bowman et al., 2015) is a dataset of 570k human-written sentence pairs manually labeled with the labels entailment, contradiction, and neutral. The source for the premise sentences in SNLI were image captions taken from the Flickr30k corpus (Young et al., 2014).

MultiNLI

The Multi-Genre Natural Language Inference (MultiNLI) corpus (Williams et al., 2018) consisting of 433k human-written sentence pairs labeled with entailment, contradiction and neutral. MultiNLI contains sentence pairs from ten distinct genres of both written and spoken English. Only five genres are included in the training set. The development and test sets have been divided into matched and mismatched, where the former includes only sentences from the same genres as the training data, and the latter includes sentences from the remaining genres not present in the training data.

We used the matched development set (MultiNLI-m) for the experiments.[3] The MultiNLI dataset was annotated using very similar instructions as for the SNLI dataset.[4] Therefore we can assume that the definitions of entailment, contradiction and neutral is the same in these two datasets.

SICK

SICK (Marelli et al., 2014) is a dataset that was originally constructed to test compositional distributional semantics (DS) models. The dataset contains 9,840 examples pertaining to logical inference (negation, conjunction, disjunction, apposition, relative clauses, etc.). The dataset was automatically constructed taking pairs of sentences from a random subset of the 8K ImageFlickr data set (Young et al., 2014) and the SemEval 2012 STS MSRVideo Description dataset (Agirre et al., 2012).

3.2 Model and Training Details

We perform experiments with six high-performing models covering the sentence encoding models,

[2] Here we could as well have selected MultiNLI test data. However the selection does not impact our findings.

[3] Here the choice between MultiNLI matched and mismatched does not make a difference to our experimental setup.

[4] The reason we are not saying "exactly the same" is because in some cases, as the authors report, "the prompts that surround each premise sentence during hypothesis collection are slightly tailored to fit the genre of that premise sentence".

	entailment
SICK	*A person, who is riding a bike, is wearing gear which is black* *A biker is wearing gear which is black*
SNLI	*A young family enjoys feeling ocean waves lap at their feet.* *A family is at the beach.*
MultiNLI	*Kal tangled both of Adrin's arms, keeping the blades far away.* *Adrin's arms were tangled, keeping his blades away from Kal.*
	contradiction
SICK	*There is no man wearing a black helmet and pushing a bicycle* *One man is wearing a black helmet and pushing a bicycle*
SNLI	*A man with a tattoo on his arm staring to the side with vehicles and buildings behind him.* *A man with no tattoos is getting a massage.*
MultiNLI	*Also in Eustace Street is an information office and a cultural center for children, The Ark .* *The Ark, a cultural center for kids, is located in Joyce Street.*
	neutral
SICK	*A little girl in a green coat and a boy holding a red sled are walking in the snow* *A child is wearing a coat and is carrying a red sled near a child in a green and black coat*
SNLI	*An old man with a package poses in front of an advertisement.* *A man poses in front of an ad for beer.*
MultiNLI	*Enthusiasm for Disney's Broadway production of The Lion King dwindles.* *The broadway production of The Lion King was amazing, but audiences are getting bored.*

Table 2: Example sentence pairs from the three datasets.

cross-sentence attention models as well as fine-tuned pre-trained language models.

For sentence encoding models, we chose a simple one-layer bidirectional LSTM with max pooling (BiLSTM-max) with the hidden size of 600D per direction, used e.g. in InferSent (Conneau et al., 2017), and HBMP (Talman et al., 2018). For the other models, we have chosen ESIM (Chen et al., 2017), which includes cross-sentence attention, and KIM (Chen et al., 2018), which has cross-sentence attention and utilizes external knowledge. We also selected two model involving a pre-trained language model, namely ESIM + ELMo (Peters et al., 2018) and BERT (Devlin et al., 2019). KIM is particularly interesting in this context as it performed significantly better than other models in the Breaking NLI experiment conducted by Glockner et al. (2018). The success of pre-trained language models in multiple NLP tasks make ESIM + ELMo and BERT interesting additions to this experiment. Table 3 lists the different models used in the experiments.

For BiLSTM-max we used the Adam optimizer (Kingma and Ba, 2015), a learning rate of 5e-4 and batch size of 64. The learning rate was decreased by the factor of 0.2 after each epoch if the model did not improve. Dropout of 0.1 was used between the layers of the multi-layer perceptron classifier, except before the last layer.The BiLSTM-max models were initialized with pre-trained GloVe 840B word embeddings of size 300 dimensions (Pennington et al., 2014), which were fine-tuned during training. Our BiLSMT-max model was implemented in PyTorch.

For HBMP, ESIM, KIM and BERT we used the original implementations with the default settings and hyperparameter values as described in Talman et al. (2018), Chen et al. (2017), Chen et al. (2018) and Devlin et al. (2019) respectively. For BERT we used the uncased 768-dimensional model (BERT-base). For ESIM + ELMo we used the AllenNLP (Gardner et al., 2018) PyTorch implementation with the default settings and hyperparameter values.

4 Experimental Results

Table 4 contains all the experimental results.

Our experiments show that, while all of the six models perform well when the test set is drawn from the same corpus as the training and development set, accuracy is significantly lower when we test these trained models on a test set drawn from a separate NLI corpus, the average difference in accuracy being 24.9 points across all experiments.

Accuracy drops the most when a model is tested on SICK. The difference in this case is between 19.0-29.0 points when trained on MultiNLI, between 31.6-33.7 points when trained on SNLI and between 31.1-33.0 when trained on SNLI + MultiNLI. This was expected, as the method of constructing the sentence pairs was different, and hence there is too much difference in the kind of sentence pairs included in the training and test sets for transfer learning to work. However, the drop

Model	Model type
BiLSTM-max (Conneau et al., 2017)	Sentence encoding
HBMP (Talman et al., 2018)	Sentence encoding
ESIM (Chen et al., 2017)	Cross-sentence attention
KIM (Chen et al., 2018)	Cross-sentence attention
ESIM + ELMo (Peters et al., 2018)	Pre-trained language model
BERT-base (Devlin et al., 2019)	Cross-sentence attention + pre-trained language model

Table 3: Model architectures used in the experiments.

was more dramatic than expected.

The most surprising result was that the accuracy of all models drops significantly even when the models were trained on MultiNLI and tested on SNLI (3.6-11.1 points). This is surprising as both of these datasets have been constructed with a similar data collection method using the same definition of entailment, contradiction and neutral. The sentences included in SNLI are also much simpler compared to those in MultiNLI, as they are taken from the Flickr image captions. This might also explain why the difference in accuracy for all of the six models is lowest when the models are trained on MultiNLI and tested on SNLI. It is also very surprising that the model with the biggest difference in accuracy was ESIM + ELMo which includes a pre-trained ELMo language model. BERT performed significantly better than the other models in this experiment having an accuracy of 80.4% and only 3.6 point difference in accuracy.

The poor performance of most of the models with the MultiNLI-SNLI dataset pair is also very surprising given that neural network models do not seem to suffer a lot from introduction of new genres to the test set which were not included in the training set, as can be seen from the small difference in test accuracies for the matched and mismatched test sets (see e.g Williams et al., 2018). In a sense SNLI could be seen as a separate genre not included in MultiNLI. This raises the question if the SNLI and MultiNLI have e.g. different kinds of annotation artifacts, which makes transfer learning between these datasets more difficult.

All the models, except BERT, perform almost equally poorly across all the experiments. Both BiLSTM-max and HBMP have an average drop in accuracy of 24.4 points, while the average for KIM is 25.5 and for ESIM + ELMo 25.6. ESIM has the highest average difference of 27.0 points. In contrast to the findings of Glockner et al. (2018), utilizing external knowledge did not improve the model's generalization capability, as KIM performed equally poorly across all dataset combinations.

Also including a pretrained ELMo language model did not improve the results significantly. The overall performance of BERT was significantly better than the other models, having the lowest average difference in accuracy of 22.5 points. Our baselines for SNLI (90.4%) and SNLI + MultiNLI (90.6%) outperform the previous state-of-the-art accuracy for SNLI (90.1%) by Kim et al. (2018).

To understand better the types of errors made by neural network models in NLI we looked at some example failure-pairs for selected models.[5] Tables 5 and 6 contain some randomly selected failure-pairs for two models: BERT and HBMP, and for three set-ups: SNLI→SICK, SNLI→MultiNLI and MultiNLI→SICK. We chose BERT as the current the state of the art NLI model. HBMP was selected as a high performing model in the sentence encoding model type. Although the listed sentence pairs represent just a small sample of the errors made by these models, they do include some interesting examples. First, it seems that SICK has a more narrow notion of contradiction – corresponding more to logical contradiction – compared to the contradiction in SNLI and MultiNLI, where especially in SNLI the sentences are contradictory if they describe a different state of affairs. This is evident in the sentence pair: *A young child is running outside over the fallen leaves* and *A young child is lying down on a gravel road that is covered with dead leaves*, which is predicted by BERT to be *contradiction* although the gold label is *neutral*. Another interesting example is the sentence pair: *A boat pear with people boarding and*

[5] More thorough error analysis of each of the models and set-up is out of scope of this work but we intend to address these in our future research.

Train data	Test data	Test accuracy	Δ	Model
SNLI	**SNLI**	**86.1**		**BiLSTM-max** (our baseline)
SNLI	**SNLI**	**86.6**		**HBMP** (Talman et al., 2018)
SNLI	**SNLI**	**88.0**		**ESIM** (Chen et al., 2017)
SNLI	**SNLI**	**88.6**		**KIM** (Chen et al., 2018)
SNLI	**SNLI**	**88.6**		**ESIM + ELMo** (Peters et al., 2018)
SNLI	**SNLI**	**90.4**		**BERT-base** (Devlin et al., 2019)
SNLI	MultiNLI-m	55.7*	-30.4	BiLSTM-max
SNLI	MultiNLI-m	56.3*	-30.3	HBMP
SNLI	MultiNLI-m	59.2*	-28.8	ESIM
SNLI	MultiNLI-m	61.7*	-26.9	KIM
SNLI	MultiNLI-m	64.2*	-24.4	ESIM + ELMo
SNLI	MultiNLI-m	75.5*	-14.9	BERT-base
SNLI	SICK	54.5	-31.6	BiLSTM-max
SNLI	SICK	53.1	-33.5	HBMP
SNLI	SICK	54.3	-33.7	ESIM
SNLI	SICK	55.8	-32.8	KIM
SNLI	SICK	56.7	-31.9	ESIM + ELMo
SNLI	SICK	56.9	-33.5	BERT-base
MultiNLI	**MultiNLI-m**	**73.1***		**BiLSTM-max**
MultiNLI	**MultiNLI-m**	**73.2***		**HBMP**
MultiNLI	**MultiNLI-m**	**76.8***		**ESIM**
MultiNLI	**MultiNLI-m**	**77.3***		**KIM**
MultiNLI	**MultiNLI-m**	**80.2***		**ESIM + ELMo**
MultiNLI	**MultiNLI-m**	**84.0***		**BERT-base**
MultiNLI	SNLI	63.8	-9.3	BiLSTM-max
MultiNLI	SNLI	65.3	-7.9	HBMP
MultiNLI	SNLI	66.4	-10.4	ESIM
MultiNLI	SNLI	68.5	-8.8	KIM
MultiNLI	SNLI	69.1	-11.1	ESIM + ELMo
MultiNLI	SNLI	80.4	-3.6	BERT-base
MultiNLI	SICK	54.1	-19.0	BiLSTM-max
MultiNLI	SICK	54.1	-19.1	HBMP
MultiNLI	SICK	47.9	-28.9	ESIM
MultiNLI	SICK	50.9	-26.4	KIM
MultiNLI	SICK	51.4	-28.8	ESIM + ELMo
MultiNLI	SICK	55.0	-29.0	BERT-base
SNLI + MultiNLI	**SNLI**	**86.1**		**BiLSTM-max**
SNLI + MultiNLI	**SNLI**	**86.1**		**HBMP**
SNLI + MultiNLI	**SNLI**	**87.5**		**ESIM**
SNLI + MultiNLI	**SNLI**	**86.2**		**KIM**
SNLI + MultiNLI	**SNLI**	**88.8**		**ESIM + ELMo**
SNLI + MultiNLI	**SNLI**	**90.6**		**BERT-base**
SNLI + MultiNLI	SICK	54.5	-31.6	BiLSTM-max
SNLI + MultiNLI	SICK	55.0	-31.1	HBMP
SNLI + MultiNLI	SICK	54.5	-33.0	ESIM
SNLI + MultiNLI	SICK	54.6	-31.6	KIM
SNLI + MultiNLI	SICK	57.1	-31.7	ESIM + ELMo
SNLI + MultiNLI	SICK	59.1	-31.5	BERT-base

Table 4: Test accuracies (%). For the baseline results (highlighted in bold) the training data and test data have been drawn from the same benchmark corpus. Δ is the difference between the test accuracy and the baseline accuracy for the same training set. Results marked with * are for the development set, as no annotated test set is openly available. Best scores with respect to accuracy and difference in accuracy are underlined.

disembarking some boats. and *people are boarding and disembarking some boats*, which is incorrectly predicted by BERT to be *contradiction* although it has been labeled as *entailment*. Here the two sentences describe the same event from different points of view: the first one describing a boat pear with some people on it and the second one describing the people directly. Interestingly the added information about the boat pear seems to confuse the model.

5 Discussion and Conclusion

In this paper we have shown that neural network models for NLI fail to generalize across different NLI benchmarks. We experimented with six state-of-the-art models covering sentence en-

coding approaches, cross-sentence attention models and pre-trained and fine-tuned language models. For all the systems, the accuracy drops between 3.6-33.7 points (the average drop being 24.9 points), when testing with a test set drawn from a separate corpus from that of the training data, as compared to when the test and training data are splits from the same corpus. Our findings, together with the previous negative findings, indicate that the state-of-the-art models fail to capture the semantics of NLI in a way that will enable them to generalize across different NLI situations.

The results highlight two issues to be taken into consideration: a) using datasets involving a fraction of what NLI is, will fail when tested in datasets that are testing for a slightly different definition of inference. This is evident when we move from the SNLI to the SICK dataset. b) NLI is to some extent genre/context dependent. Training on SNLI and testing on MultiNLI gives worse results than vice versa. This is particularly evident in the case of BERT. These results highlight that training on multiple genres helps. However, this help is still not enough given that, even in the case of training on MultiNLI (multi genre) and training on SNLI (single genre and same definition of inference with MultiNLI), accuracy drops significantly.

We also found that involving a large pre-trained language model helps with transfer learning when the datasets are similar enough, as is the case with SNLI and MultiNLI. Our results further corroborate the power of pre-trained and fine-tuned language models like BERT in NLI. However, not even BERT is able to generalize from SNLI and MultiNLI to SICK, possibly due to the difference between what kind of inference relations are contained in these datasets.

Our findings motivate us to look for novel neural network architectures and approaches that better capture the semantics on natural language inference beyond individual datasets. However, there seems to be a need to start with better constructed datasets, i.e. datasets that will not only capture fractions of what NLI is in reality. Better NLI systems need to be able to be more versatile on the types of inference they can recognize. Otherwise, we would be stuck with systems that can cover only some aspects of NLI. On a theoretical level, and in connection to the previous point, we need a better understanding of the range of phenomena NLI must be able to cover and focus our future endeavours for dataset construction towards this direction. In order to do this a more systematic study is needed on the different kinds of entailment relations NLI datasets need to include. Our future work will include a more systematic and broad-coverage analysis of the types of errors the models make and in what kinds of sentence-pairs they make successful predictions.

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BERT: SNLI → SICK	
A young child is running outside over the fallen leaves A young child is lying down on a gravel road that is covered with dead leaves	
Label: neutral	**Prediction:** contradiction
The man is being knocked off of a horse Someone is falling off a horse	
Label: entailment	**Prediction:** contradiction
There is no one typing Someone is typing on a keyboard	
Label: contradiction	**Prediction:** neutral
The man in the purple hat is operating a camera that makes videos There is no man with a camera studying the subject	
Label: neutral	**Prediction:** contradiction
A woman is taking off a cloak, which is very large, and revealing an extravagant dress A woman is putting on a cloak, which is very large, and concealing an extravagant dress	
Label: contradiction	**Prediction:** neutral
BERT: MultiNLI → SICK	
A cowboy is riding a horse and cornering a barrel A cowgirl is riding a horse and corners a barrel	
Label: neutral	**Prediction:** contradiction
A tan dog is jumping up and catching a tennis ball A dog with a tan coat is jumping up and catching a tennis ball	
Label: entailment	**Prediction:** neutral
The bunch of men are playing rugby on a muddy field Some men are idling	
Label: neutral	**Prediction:** contradiction
A blond child is going down a slide and throwing up his arms A child with dark hair is going down a slide and throwing up his arms	
Label: contradiction	**Prediction:** entailment
There is no person in bike gear standing steadily in front of the mountains A group of people is equipped with protective gear	
Label: neutral	**Prediction:** contradiction
BERT: MultiNLI → SNLI	
A woman in a white wedding dress is being dressed and fitted by two other women. A woman is being fitted for the first time.	
Label: neutral	**Prediction:** contradiction
A boat pear with people boarding and disembarking some boats. people are boarding and disembarking some boats	
Label: entailment	**Prediction:** contradiction
Several men at a bar watch a sports game on the television. The men are at a baseball game.	
Label: contradiction	**Prediction:** entailment
A singer wearing a leather jacket performs on stage with dramatic lighting behind him. a singer is on american idol	
Label: neutral	**Prediction:** contradiction
A person rolls down a hill riding a wagon as another watches. A person stares at an empty hill.	
Label: contradiction	**Prediction:** neutral

Table 5: Example failure-pairs for BERT.

HBMP: SNLI → SICK
a boy is sitting in a room and playing a piano by lamp light a boy is playing a keyboard
Label: entailment **Prediction:** contradiction
a woman is wearing ear protection and is firing a gun at an outdoor shooting range a woman is shooting at target practices
Label: entailment **Prediction:** neutral
a man in a purple hat isn't climbing a rocky wall with bare hands a man in a purple hat is climbing a rocky wall with bare hands
Label: contradiction **Prediction:** entailment
a cat is swinging on a fan a cat is stuck on a moving ceiling fan
Label: neutral **Prediction:** contradiction
a young boy is jumping and covering nearby wooden fence with grass a young boy covered in grass is jumping near a wooden fence
Label: neutral **Prediction:** entailment
HBMP: MultiNLI → SICK
two dogs are walking slowly through a park two dogs are running quickly through a park
Label: neutral **Prediction:** entailment
the woman is playing a guitar which is electric the woman is playing an electric guitar
Label: entailment **Prediction:** neutral
there is no man squatting in brush and taking a photograph a man is crouching and holding a camera
Label: neutral **Prediction:** contradiction
the snowboarder is jumping off a snowy hill a snowboarder is jumping off the snow
Label: neutral **Prediction:** entailment
the boy is sitting near the blue ocean the boy is wading through the blue ocean
Label: contradiction **Prediction:** neutral
HBMP: MultiNLI → SNLI
a man is holding a book standing in front of a chalkboard. a person is in a classroom teaching.
Label: entailment **Prediction:** contradiction
a woman with a pink purse walks down a crowded sidewalk. a woman escapes a from a hostile enviroment
Label: neutral **Prediction:** contradiction
a woman with a pink purse walks down a crowded sidewalk. a woman escapes a from a hostile enviroment
Label: neutral **Prediction:** contradiction
a person waterskiing in a river with a large wall in the background. a dog waterskiing in a river with a large wall in the background.
Label: contradiction **Prediction:** neutral
a man wearing a blue shirt and headphones around his neck raises his arm. a man is raising his arm to get someones attention.
Label: neutral **Prediction:** entailment

Table 6: Example failure-pairs for HBMP.